Encoding circuits of the non-binary EA-QC-QLDPC codes (Examples 1–10)

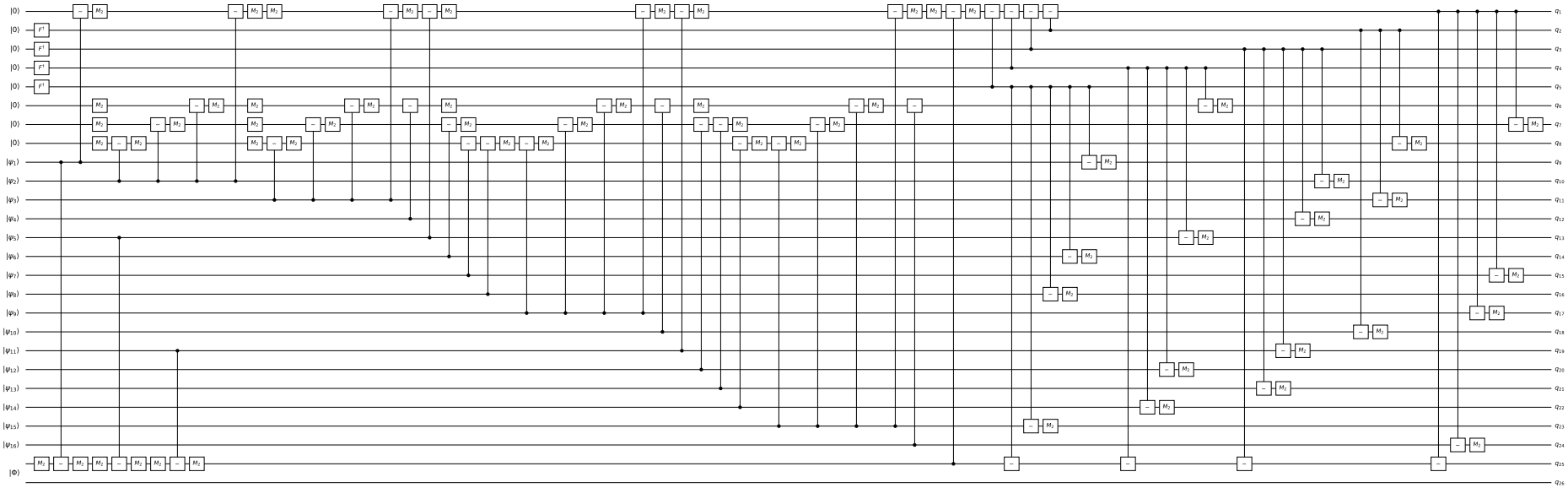


Figure 1: Encoder of the $[[25, 16; 1]]_3$ code of Example 1 obtained by the proposed encoding strategy: 59 ADD, 59 MUL, 4 DFT; two-qudit depth 21. A target labeled + (−) denotes ADD ($\text{ADD}^\dagger$), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{26} is not acted on.

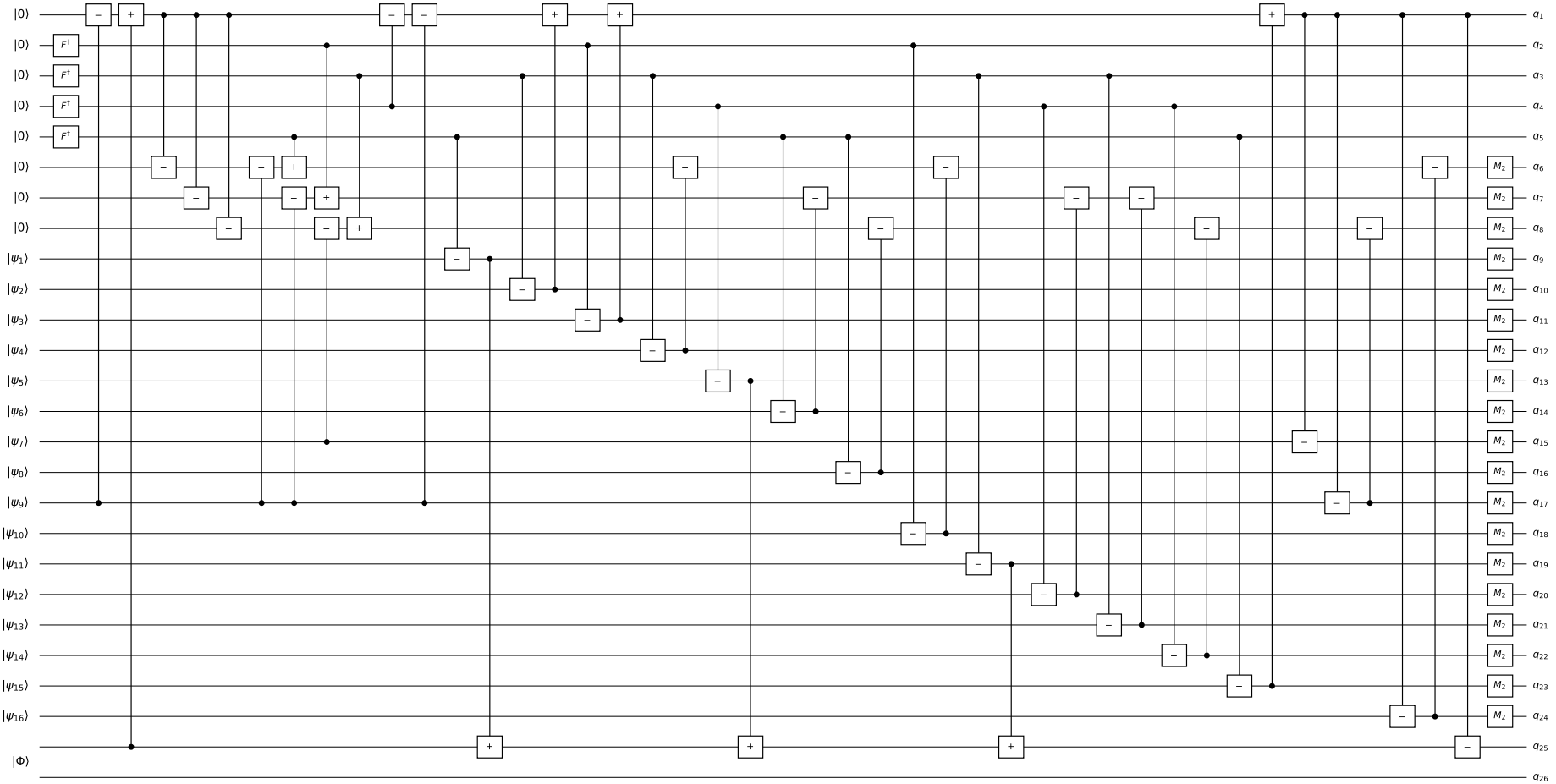


Figure 2: Encoder of the $[[25, 16; 1]]_3$ code of Example 1 obtained by the proposed encoding strategy followed by beam-search optimization: 45 ADD, 19 MUL, 4 DFT; two-qudit depth 15. A target labeled + (−) denotes ADD ($\mathrm{ADD}^\dagger$), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{26} is not acted on.

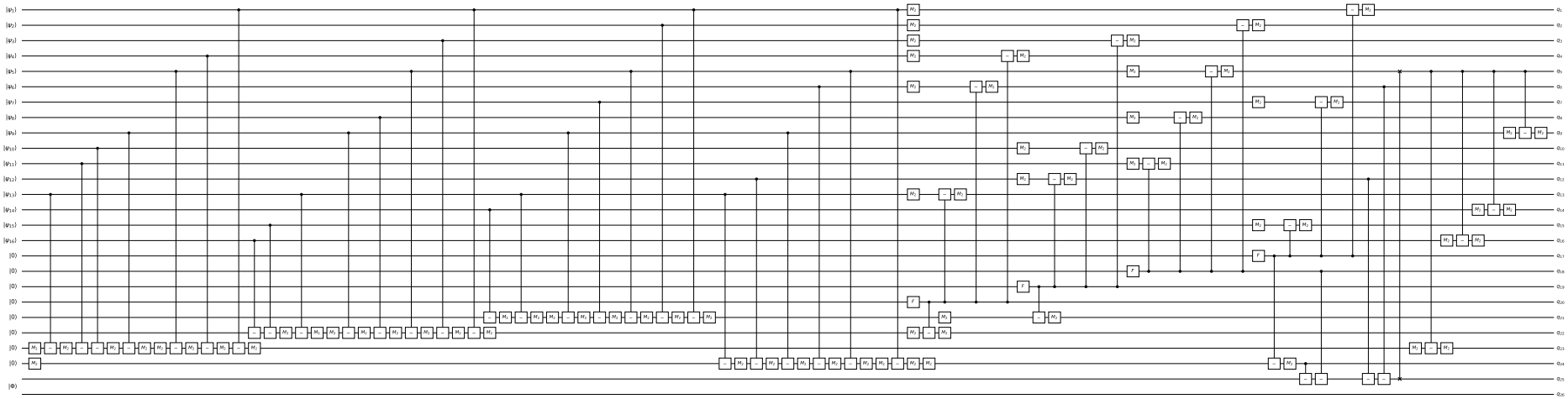


Figure 3: Encoder of the $[[25, 16; 1]]_3$ code of Example 1 obtained by the procedure of [NG]: 52 ADD, 72 MUL, 4 DFT, 1 SWAP; two-qudit depth 21. A target labeled + (−) denotes ADD ($\mathrm{ADD}^\dagger$), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{26} is not acted on.

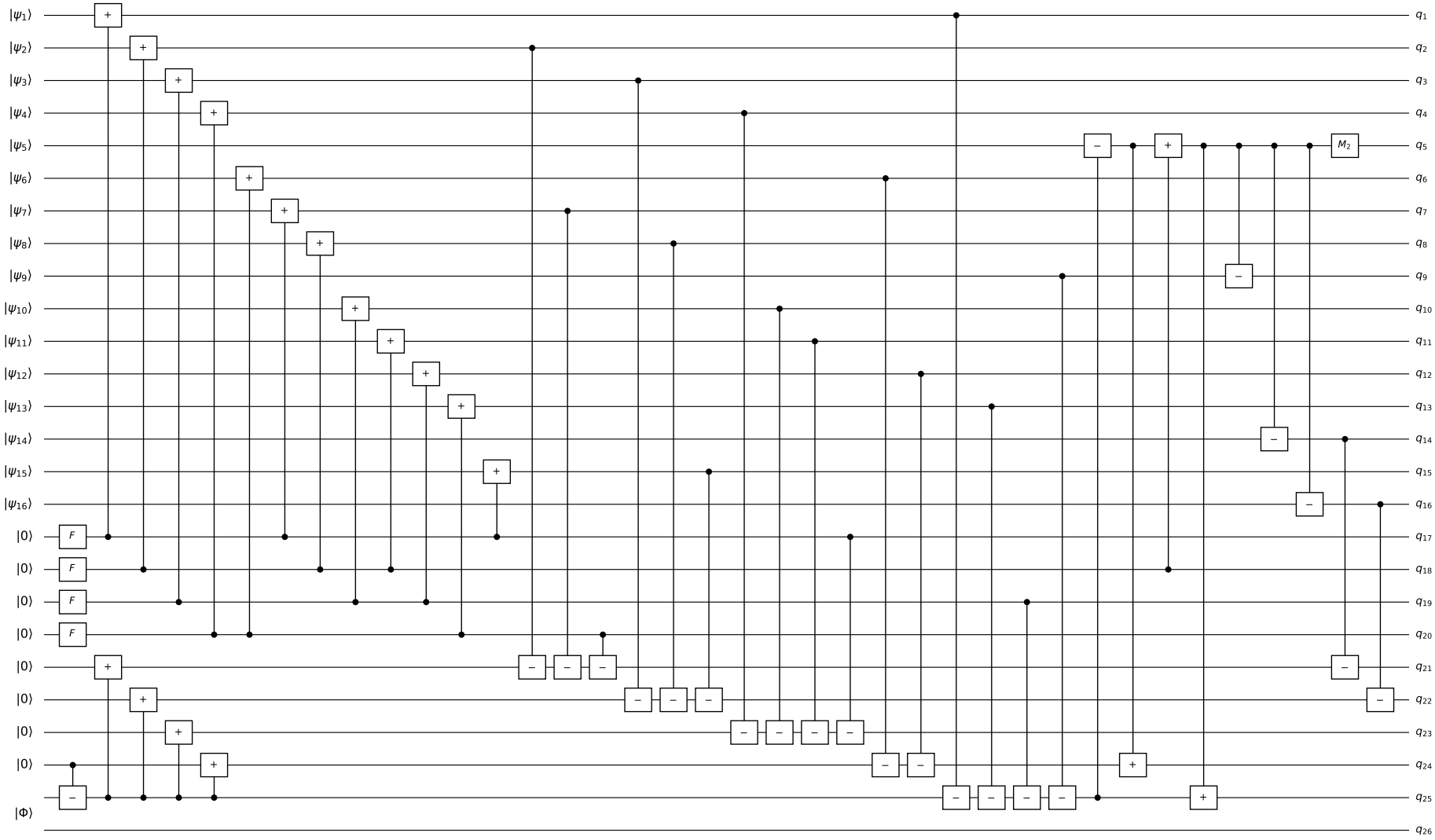


Figure 4: Encoder of the $[[25, 16; 1]]_3$ code of Example 1 obtained by the procedure of [NG] followed by beam-search optimization: 42 ADD, 1 MUL, 4 DFT; two-qudit depth 17. A target labeled + (−) denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{26} is not acted on.

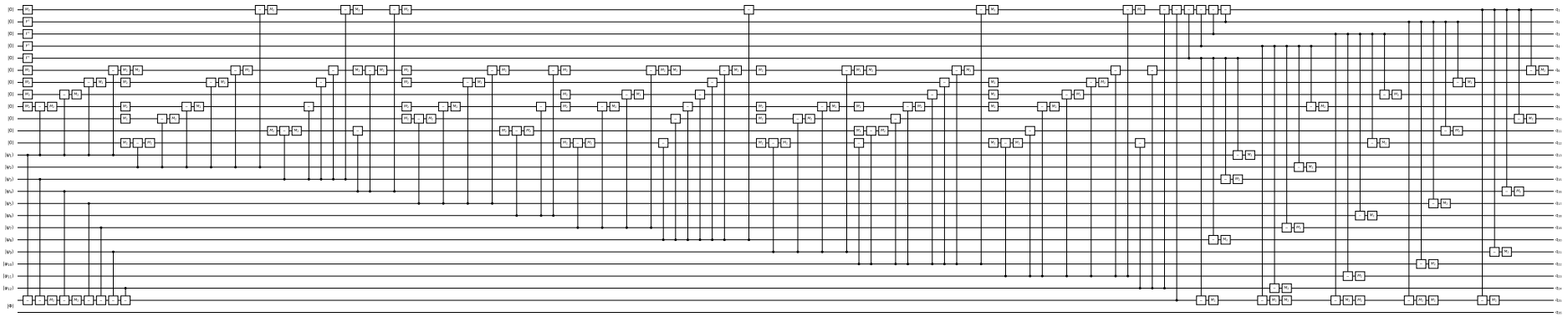


Figure 5: Encoder of the $[[25, 12; 1]]_3$ code of Example 2 obtained by the proposed encoding strategy: 94 ADD, 99 MUL, 4 DFT; two-qudit depth 29. A target labeled + (−) denotes ADD ($\text{ADD}^\dagger$), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{26} is not acted on.

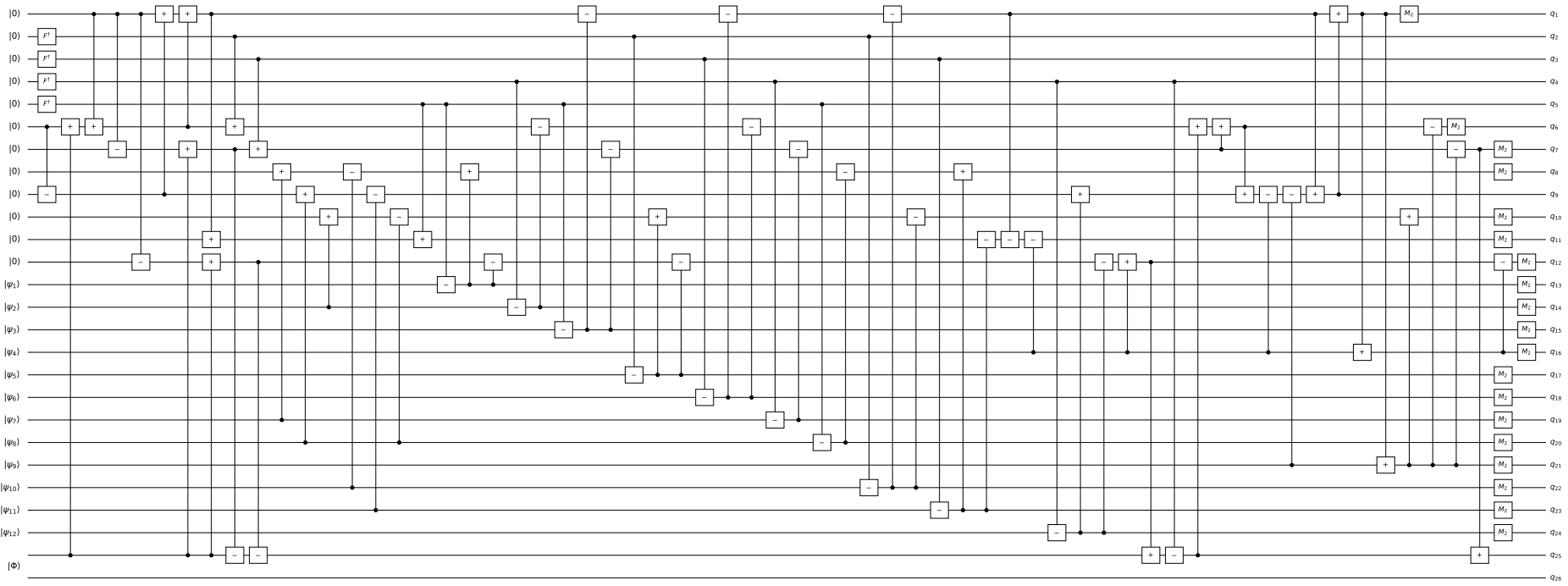


Figure 6: Encoder of the $[[25, 12; 1]]_3$ code of Example 2 obtained by the proposed encoding strategy followed by beam-search optimization: 67 ADD, 19 MUL, 4 DFT; two-qudit depth 33. A target labeled + (−) denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{26} is not acted on.

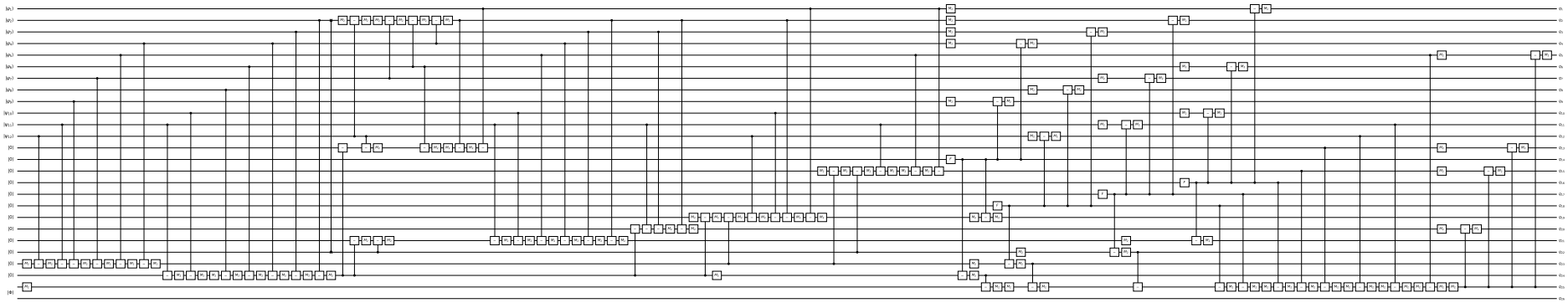


Figure 7: Encoder of the $[[25, 12; 1]]_3$ code of Example 2 obtained by the procedure of [NG]: 76 ADD, 103 MUL, 4 DFT, 1 SWAP; two-qudit depth 33. A target labeled $+$ $(-)$ denotes ADD ($\mathrm{ADD}^\dagger$), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{26} is not acted on.

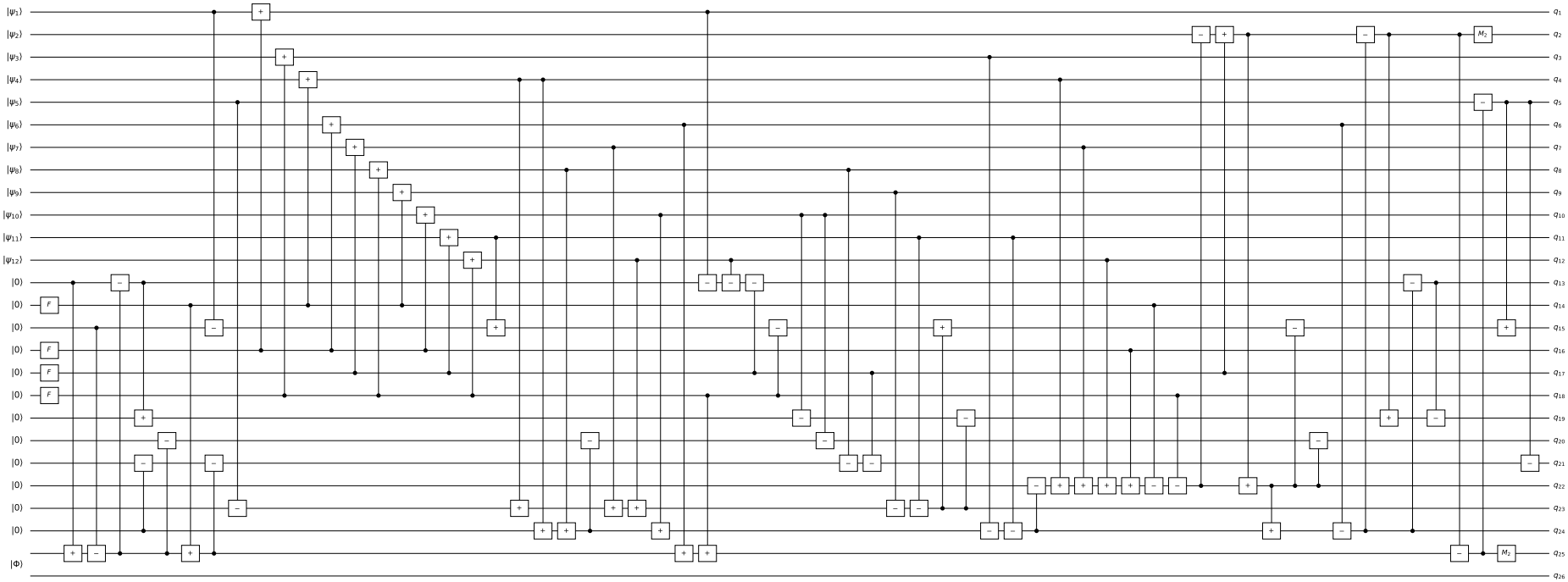


Figure 8: Encoder of the $[[25, 12; 1]]_3$ code of Example 2 obtained by the procedure of [NG] followed by beam-search optimization: 66 ADD, 2 MUL, 4 DFT; two-qudit depth 31. A target labeled + (−) denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{26} is not acted on.

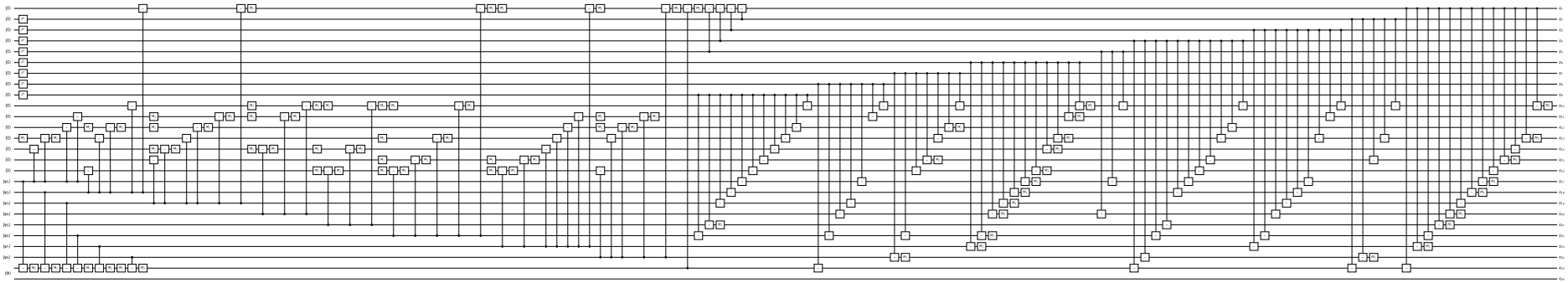


Figure 9: Encoder of the $[[25, 8; 1]]_3$ code of Example 3 obtained by the proposed encoding strategy: 126 ADD, 74 MUL, 8 DFT; two-qudit depth 34. A target labeled + (−) denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{26} is not acted on.

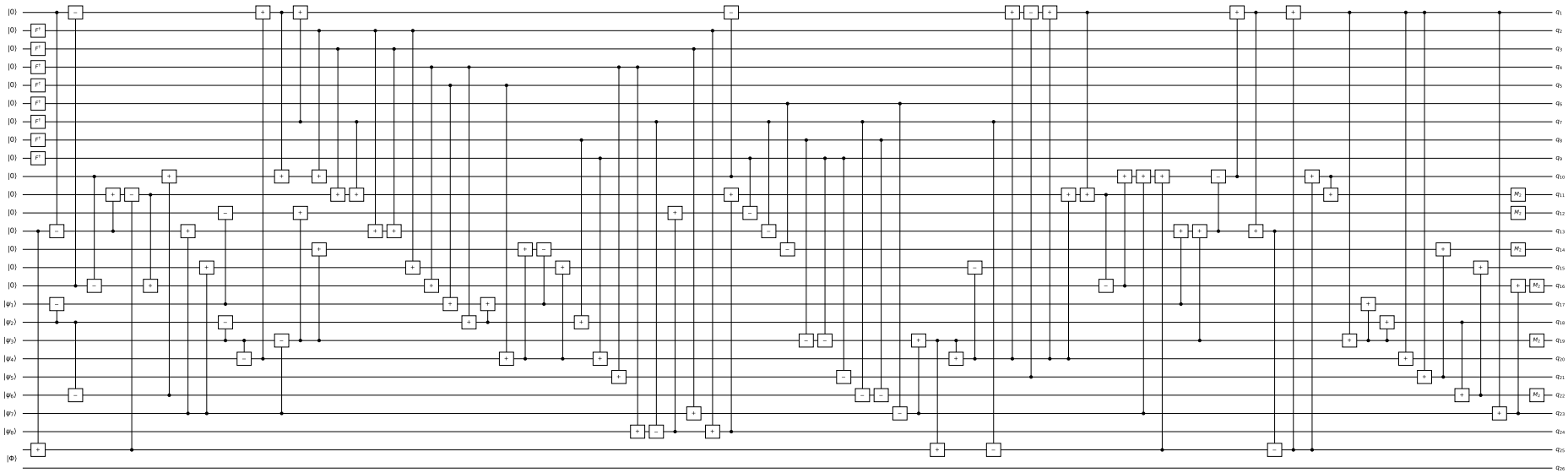


Figure 10: Encoder of the $[[25, 8; 1]]_3$ code of Example 3 obtained by the proposed encoding strategy followed by beam-search optimization: 87 ADD, 6 MUL, 8 DFT; two-qudit depth 36. A target labeled + (−) denotes ADD ($\mathrm{ADD}^\dagger$), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{26} is not acted on.

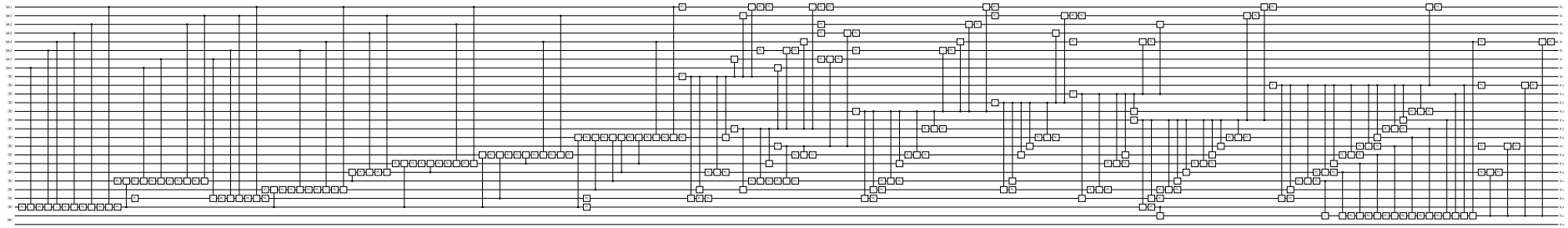


Figure 11: Encoder of the $[[25, 8; 1]]_3$ code of Example 3 obtained by the procedure of [NG]: 119 ADD, 128 MUL, 8 DFT; two-qudit depth 41. A target labeled $+$ $(-)$ denotes ADD ($\mathrm{ADD}^\dagger$), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{26} is not acted on.

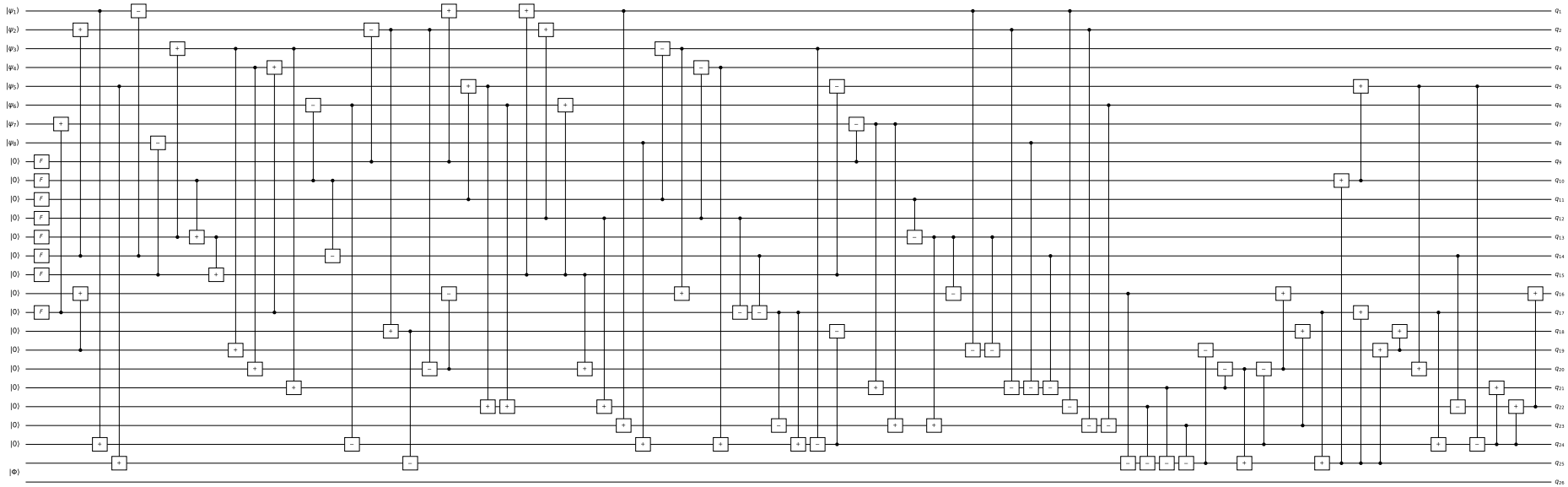


Figure 12: Encoder of the $[[25, 8; 1]]_3$ code of Example 3 obtained by the procedure of [NG] followed by beam-search optimization: 81 ADD, 0 MUL, 8 DFT; two-qudit depth 27. A target labeled + (−) denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{26} is not acted on.

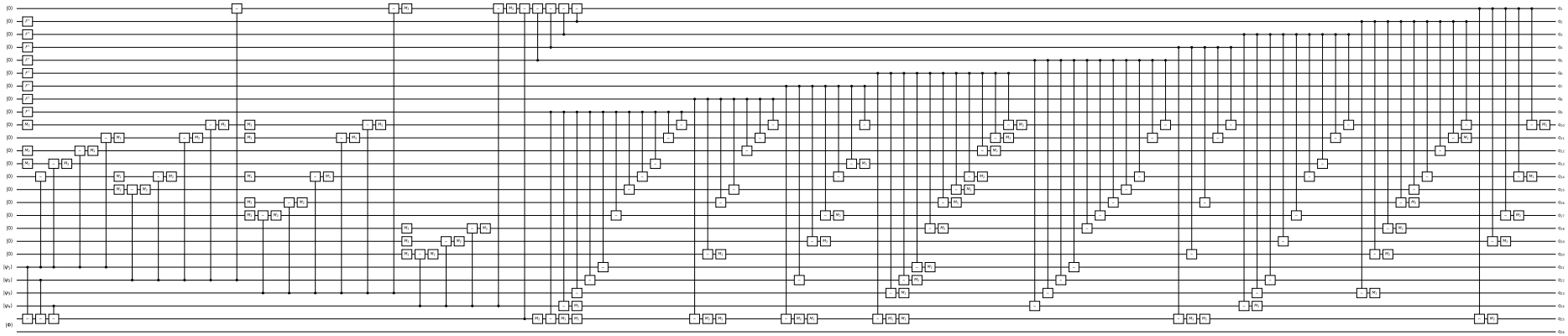


Figure 13: Encoder of the $[[25, 4; 1]]_3$ code of Example 4 obtained by the proposed encoding strategy: 102 ADD, 67 MUL, 8 DFT; two-qudit depth 30. A target labeled + (−) denotes ADD ($\mathrm{ADD}^\dagger$), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{26} is not acted on.

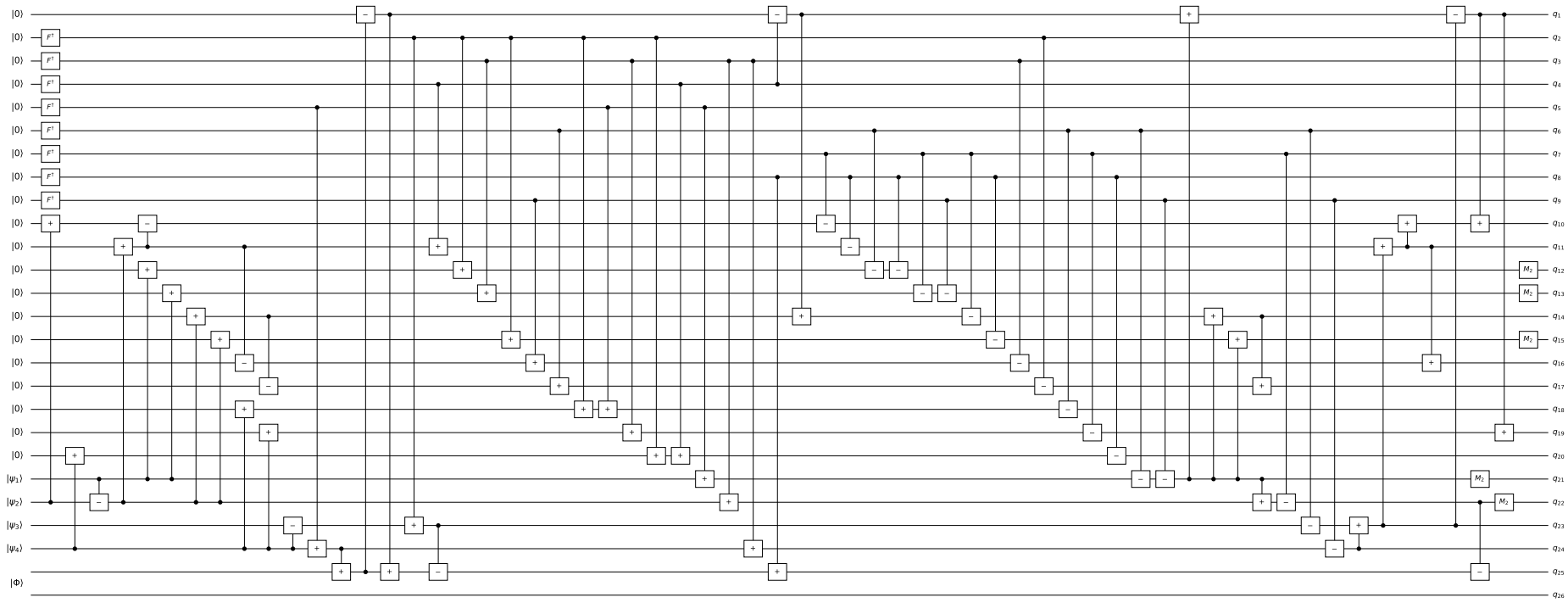


Figure 14: Encoder of the $[[25, 4; 1]]_3$ code of Example 4 obtained by the proposed encoding strategy followed by beam-search optimization: 68 ADD, 5 MUL, 8 DFT; two-qudit depth 18. A target labeled $+$ $(-)$ denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{26} is not acted on.

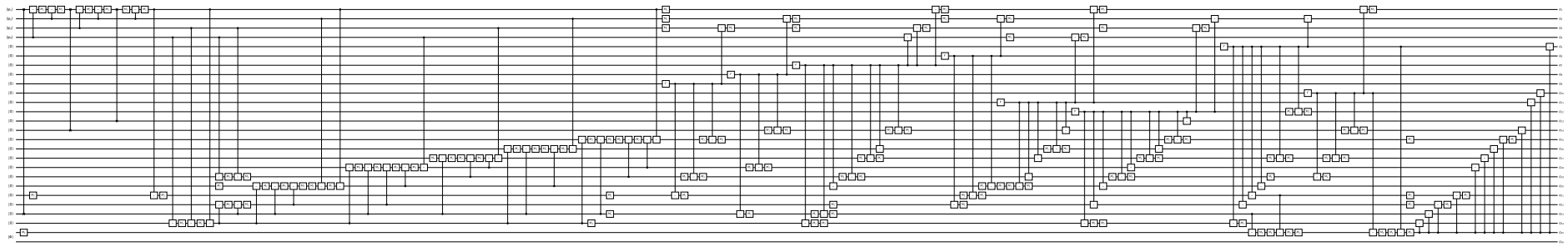


Figure 15: Encoder of the $[[25, 4; 1]]_3$ code of Example 4 obtained by the procedure of [NG]: 103 ADD, 118 MUL, 8 DFT, 3 SWAP; two-qudit depth 46. A target labeled + (−) denotes ADD ($\mathrm{ADD}^\dagger$), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{26} is not acted on.

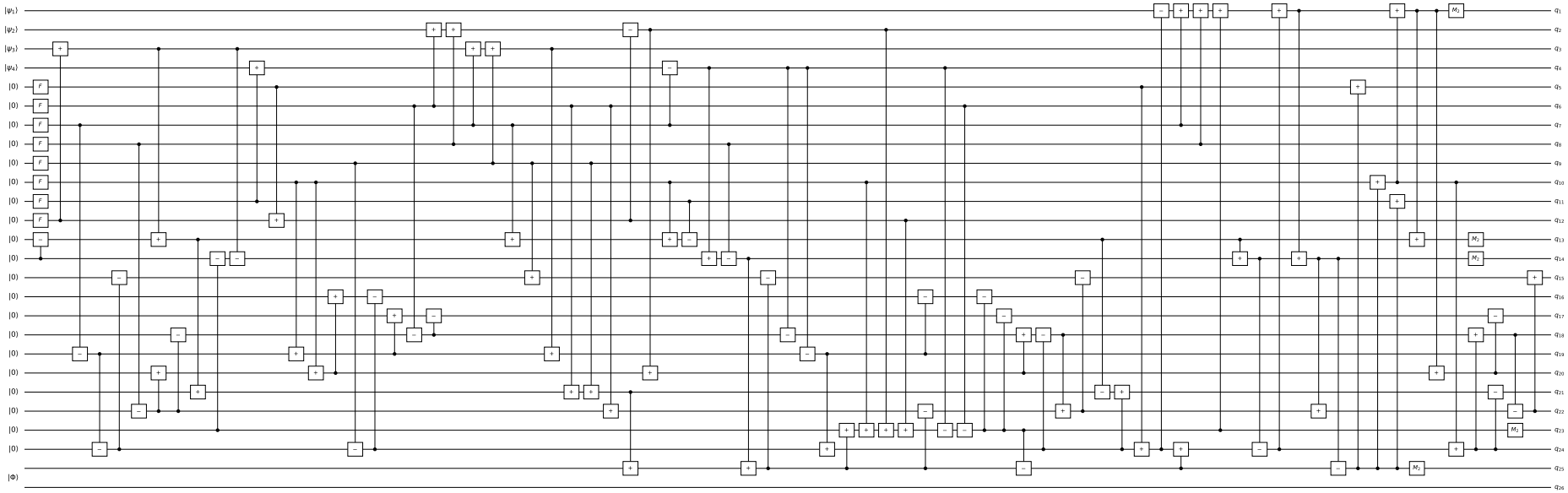


Figure 16: Encoder of the $[[25, 4; 1]]_3$ code of Example 4 obtained by the procedure of [NG] followed by beam-search optimization: 86 ADD, 5 MUL, 8 DFT; two-qudit depth 32. A target labeled $+$ ($-$) denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{26} is not acted on.

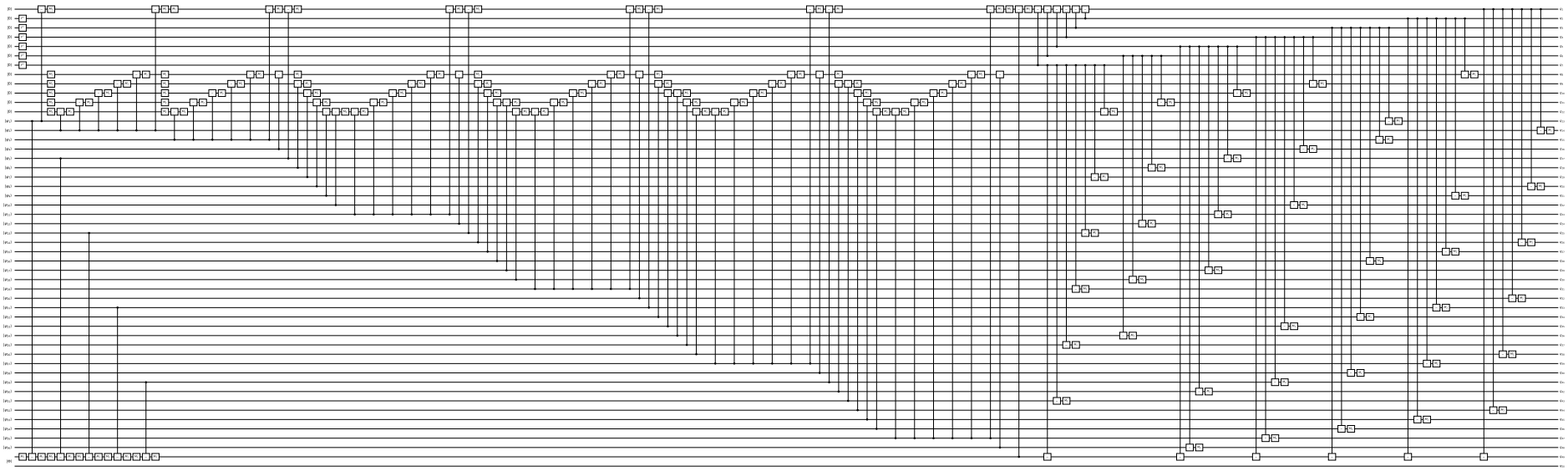


Figure 17: Encoder of the $[[49, 36; 1]]_3$ code of Example 5 obtained by the proposed encoding strategy: 125 ADD, 125 MUL, 6 DFT; two-qudit depth 34. A target labeled $+$ $(-)$ denotes ADD ($\mathrm{ADD}^\dagger$), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

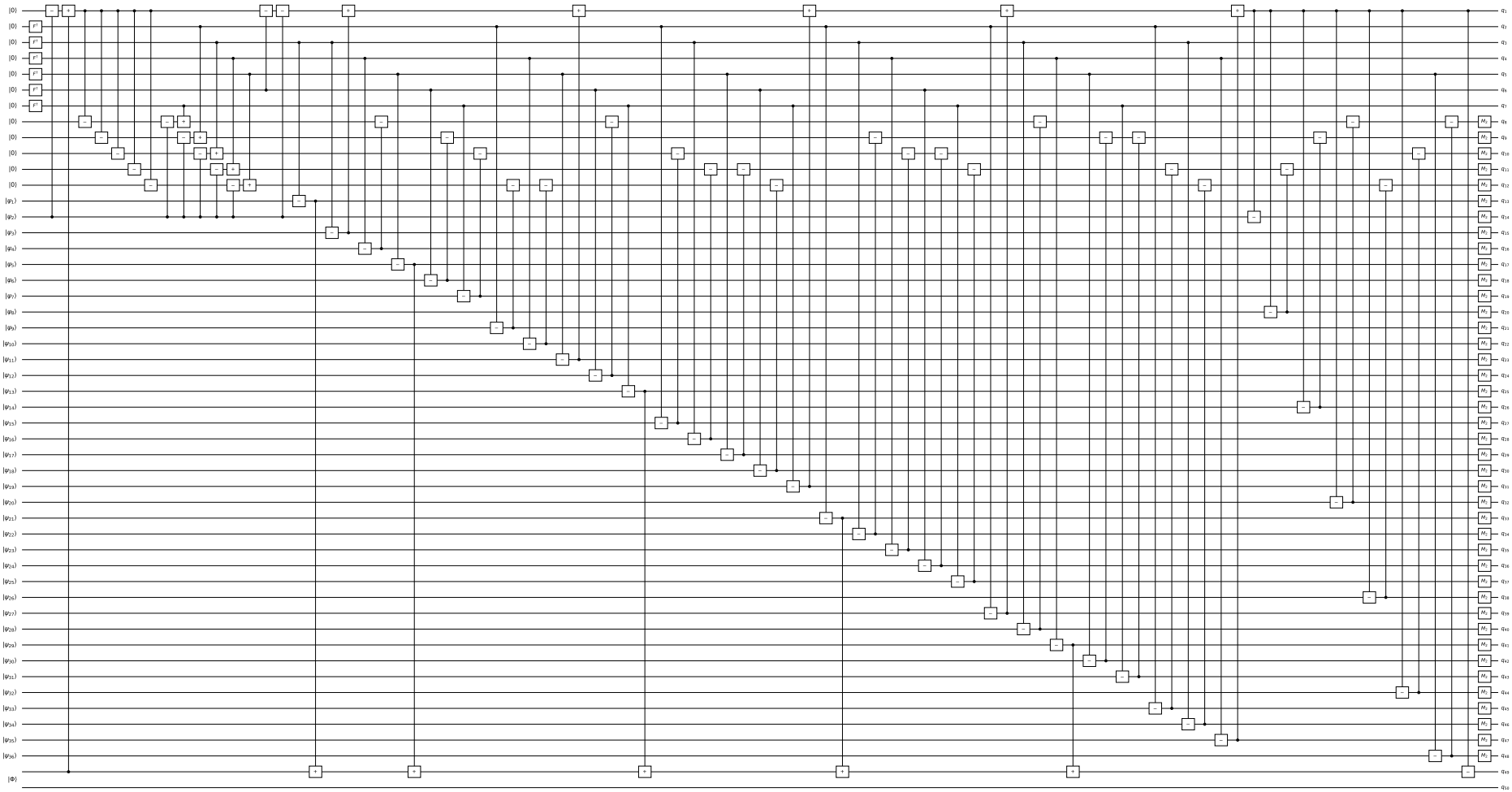


Figure 18: Encoder of the $[[49, 36; 1]]_3$ code of Example 5 obtained by the proposed encoding strategy followed by beam-search optimization: 91 ADD, 41 MUL, 6 DFT; two-qudit depth 22. A target labeled + (−) denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

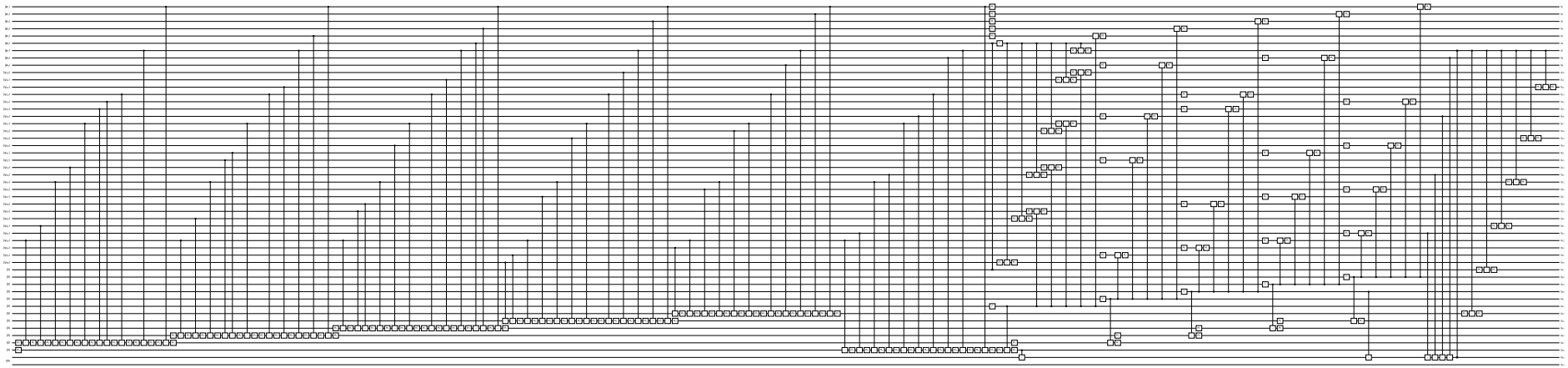


Figure 19: Encoder of the $[[49, 36; 1]]_3$ code of Example 5 obtained by the procedure of [NG]: 114 ADD, 156 MUL, 6 DFT, 2 SWAP; two-qudit depth 32. A target labeled + (−) denotes ADD ($\mathrm{ADD}^\dagger$), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

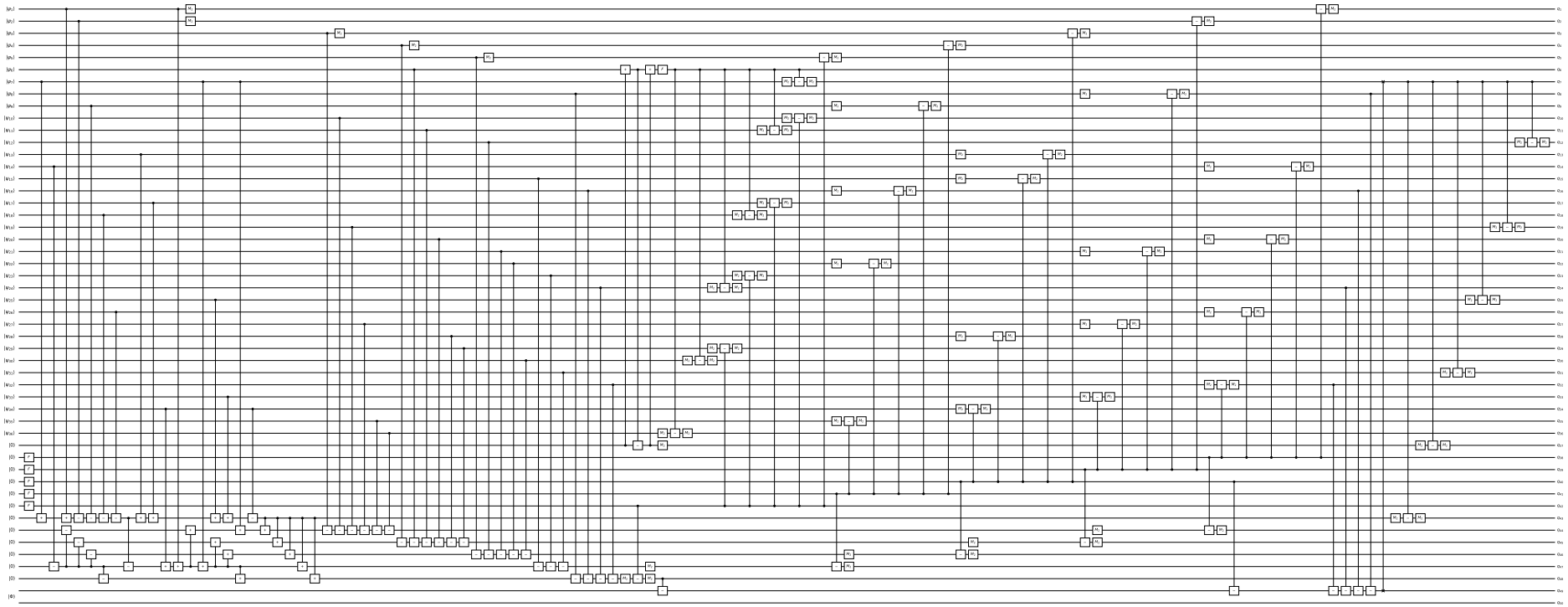


Figure 20: Encoder of the $[[49, 36; 1]]_3$ code of Example 5 obtained by the procedure of [NG] followed by beam-search optimization: 105 ADD, 85 MUL, 6 DFT, 1 SWAP; two-qudit depth 38. A target labeled + (−) denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

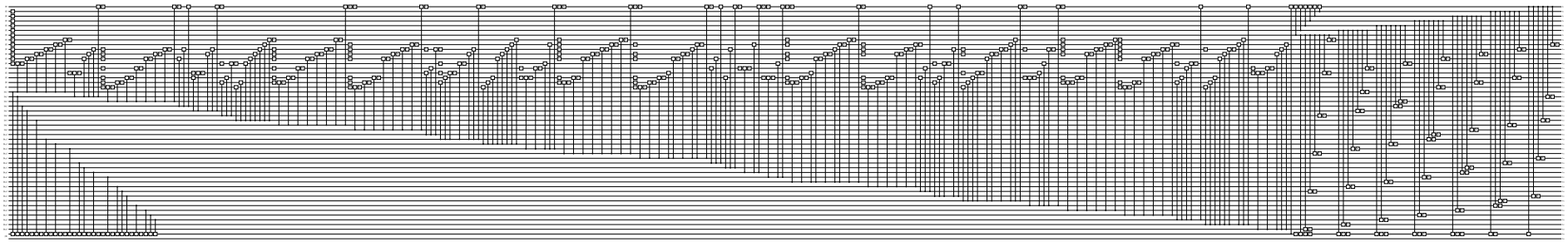


Figure 21: Encoder of the $[[49, 30; 1]]_3$ code of Example 6 obtained by the proposed encoding strategy: 249 ADD, 253 MUL, 6 DFT; two-qudit depth 48. A target labeled + (−) denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

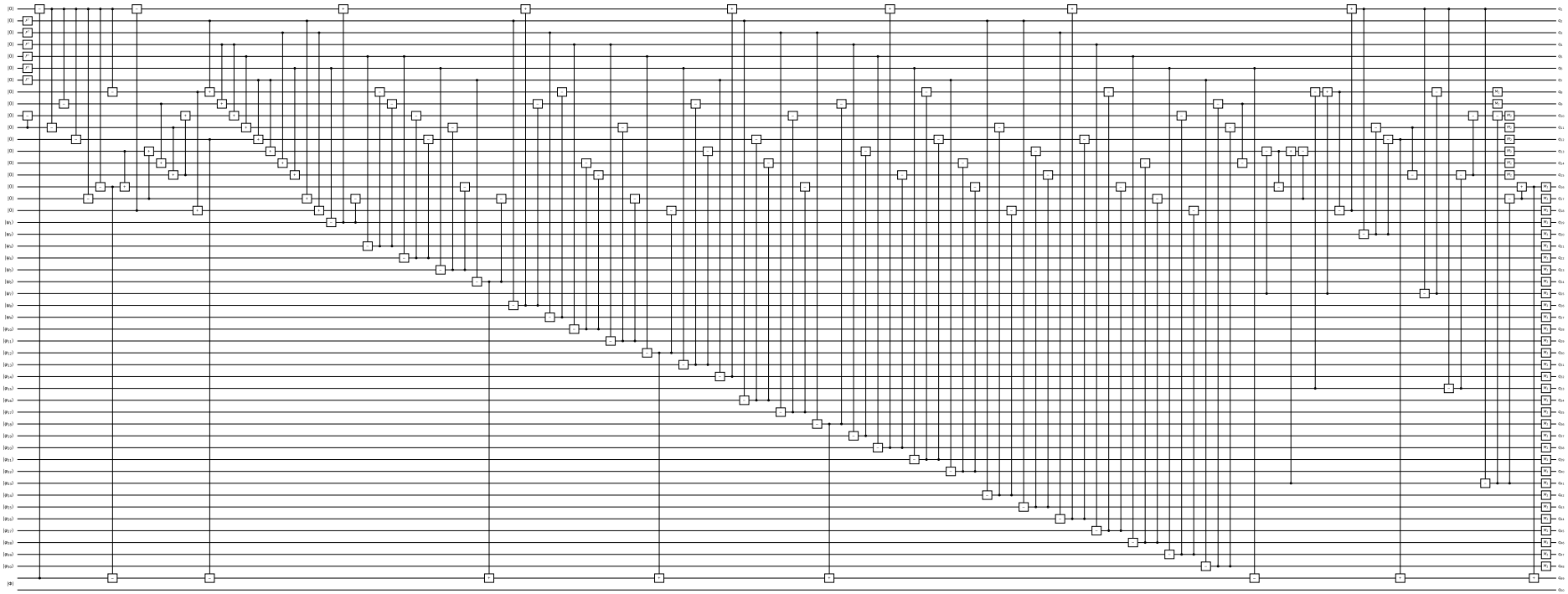


Figure 22: Encoder of the $[[49, 30; 1]]_3$ code of Example 6 obtained by the proposed encoding strategy followed by beam-search optimization: 127 ADD, 41 MUL, 6 DFT; two-qudit depth 31. A target labeled + (−) denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

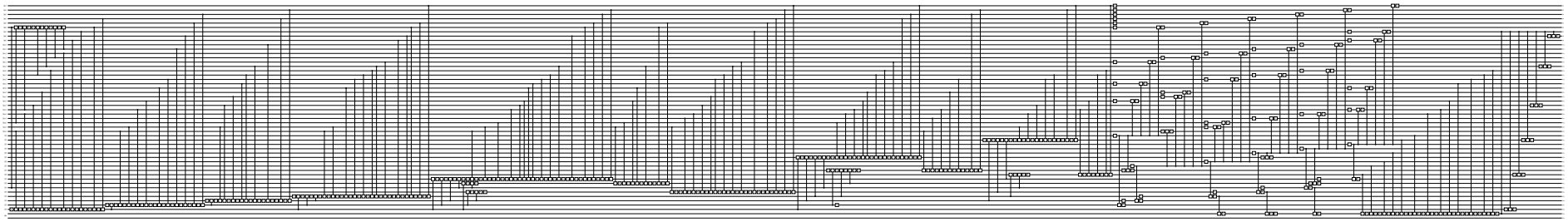


Figure 23: Encoder of the $[[49, 30; 1]]_3$ code of Example 6 obtained by the procedure of [NG]: 198 ADD, 255 MUL, 6 DFT, 2 SWAP; two-qudit depth 83. A target labeled $+$ ($-$) denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

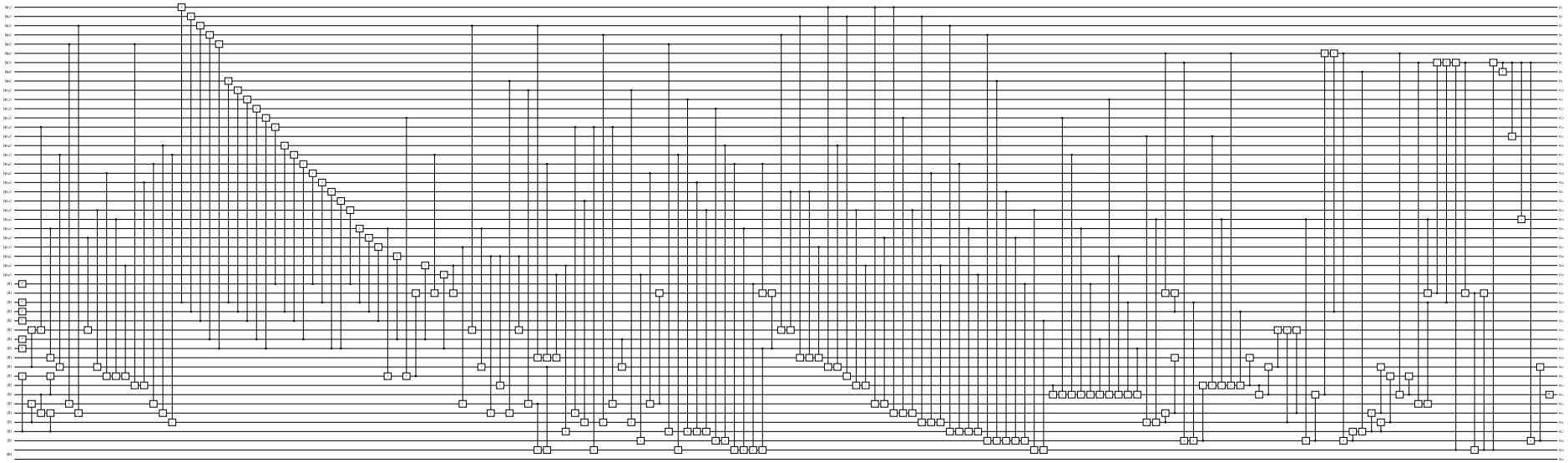


Figure 24: Encoder of the $[[49, 30; 1]]_3$ code of Example 6 obtained by the procedure of [NG] followed by beam-search optimization: 174 ADD, 1 MUL, 6 DFT; two-qudit depth 47. A target labeled $+$ ($-$) denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

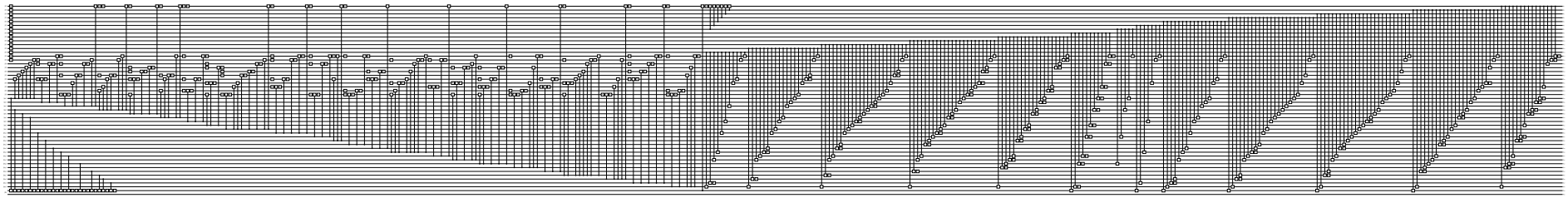


Figure 25: Encoder of the $[[49, 24; 1]]_3$ code of Example 7 obtained by the proposed encoding strategy: 357 ADD, 197 MUL, 12 DFT; two-qudit depth 61. A target labeled + (−) denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

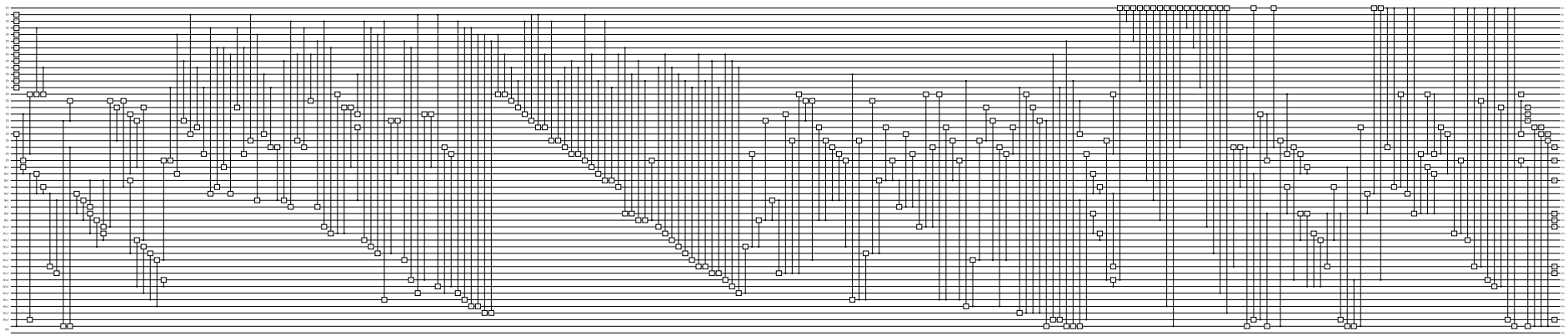


Figure 26: Encoder of the $[[49, 24; 1]]_3$ code of Example 7 obtained by the proposed encoding strategy followed by beam-search optimization: 255 ADD, 15 MUL, 12 DFT; two-qudit depth 77. A target labeled $+$ $(-)$ denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

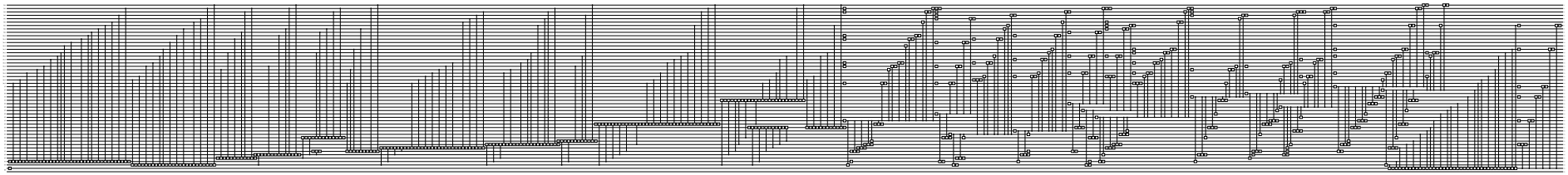


Figure 27: Encoder of the $[[49, 24; 1]]_3$ code of Example 7 obtained by the procedure of [NG]: 293 ADD, 322 MUL, 12 DFT; two-qudit depth 120. A target labeled $+$ $(-)$ denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

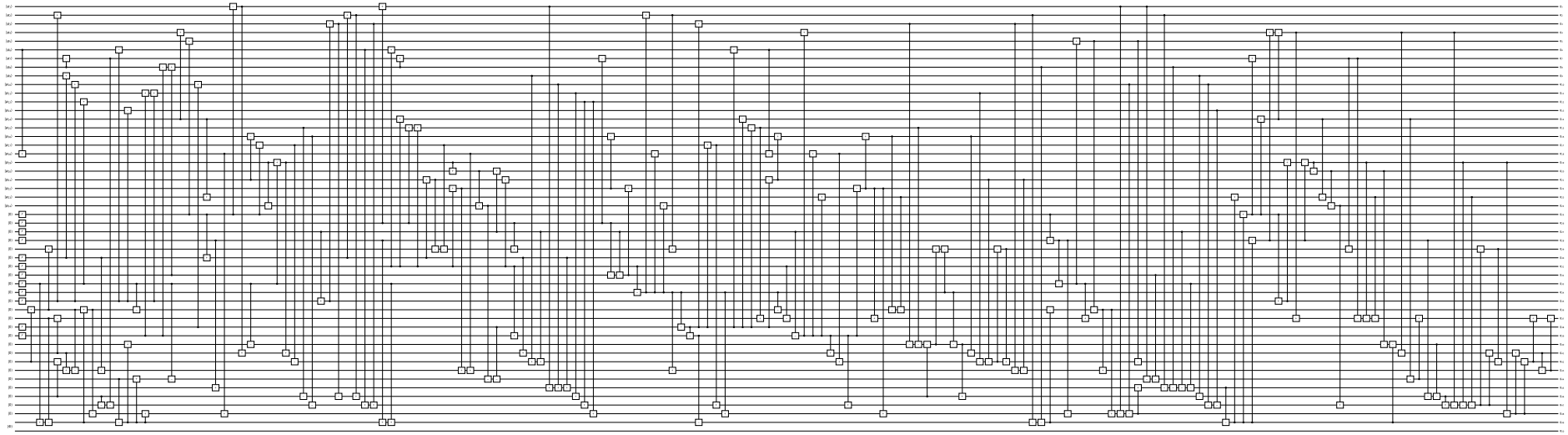


Figure 28: Encoder of the $[[49, 24; 1]]_3$ code of Example 7 obtained by the procedure of [NG] followed by beam-search optimization: 205 ADD, 0 MUL, 12 DFT; two-qudit depth 68. A target labeled + (−) denotes ADD ($\mathrm{ADD}^\dagger$), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

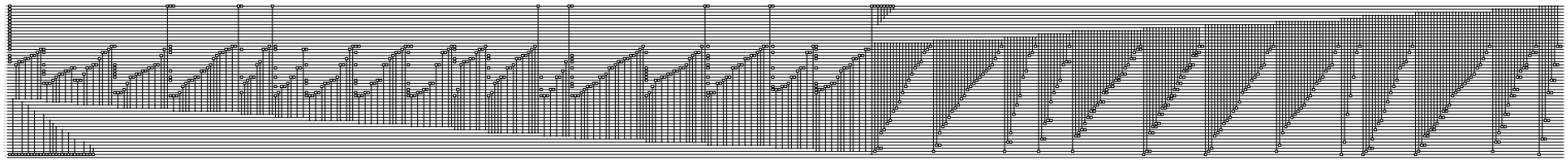


Figure 29: Encoder of the $[[49, 18; 1]]_3$ code of Example 8 obtained by the proposed encoding strategy: 420 ADD, 269 MUL, 12 DFT; two-qudit depth 66. A target labeled + (−) denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

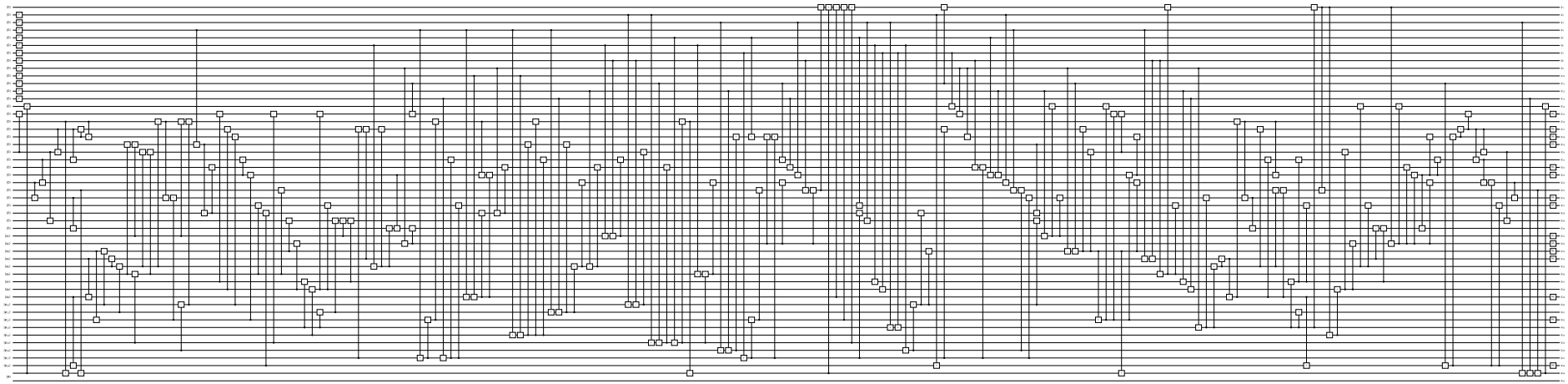


Figure 30: Encoder of the $[[49, 18; 1]]_3$ code of Example 8 obtained by the proposed encoding strategy followed by beam-search optimization: 220 ADD, 15 MUL, 12 DFT; two-qudit depth 54. A target labeled + (−) denotes ADD ($\mathrm{ADD}^\dagger$), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

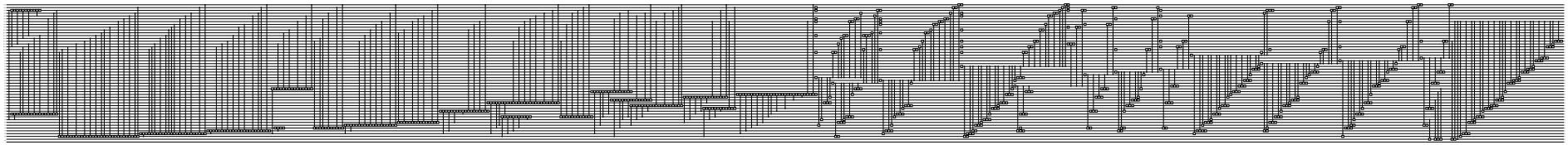


Figure 31: Encoder of the $[[49, 18; 1]]_3$ code of Example 8 obtained by the procedure of [NG]: 350 ADD, 384 MUL, 12 DFT, 2 SWAP; two-qudit depth 152. A target labeled $+$ $(-)$ denotes ADD $(\mathrm{ADD}^\dagger)$, $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

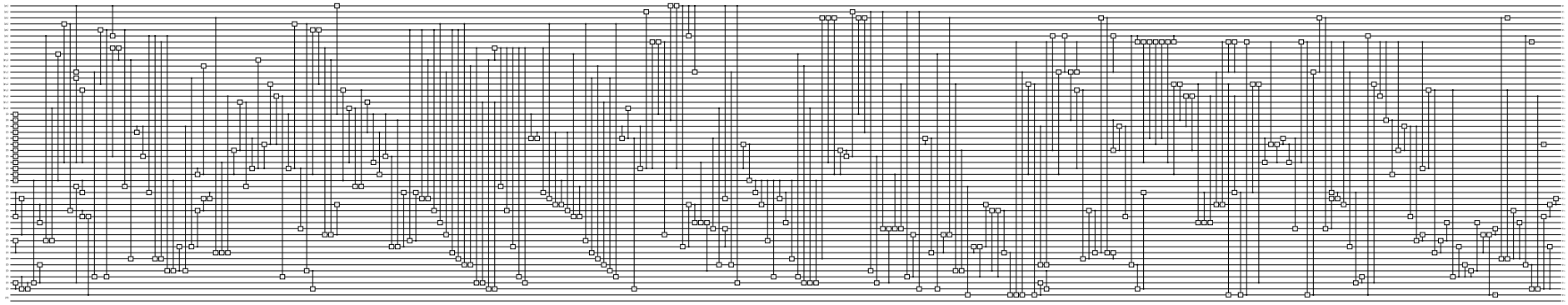


Figure 32: Encoder of the $[[49, 18; 1]]_3$ code of Example 8 obtained by the procedure of [NG] followed by beam-search optimization: 285 ADD, 4 MUL, 12 DFT; two-qudit depth 90. A target labeled + (−) denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

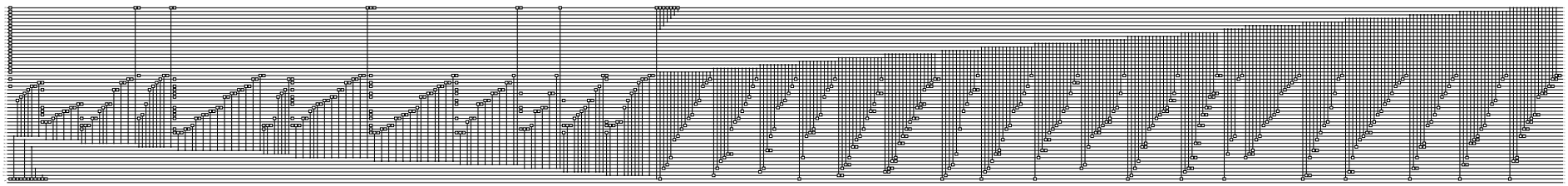


Figure 33: Encoder of the $[[49, 12; 1]]_3$ code of Example 9 obtained by the proposed encoding strategy: 375 ADD, 191 MUL, 18 DFT; two-qudit depth 61. A target labeled $+$ $(-)$ denotes ADD $(\mathrm{ADD}^\dagger)$, $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

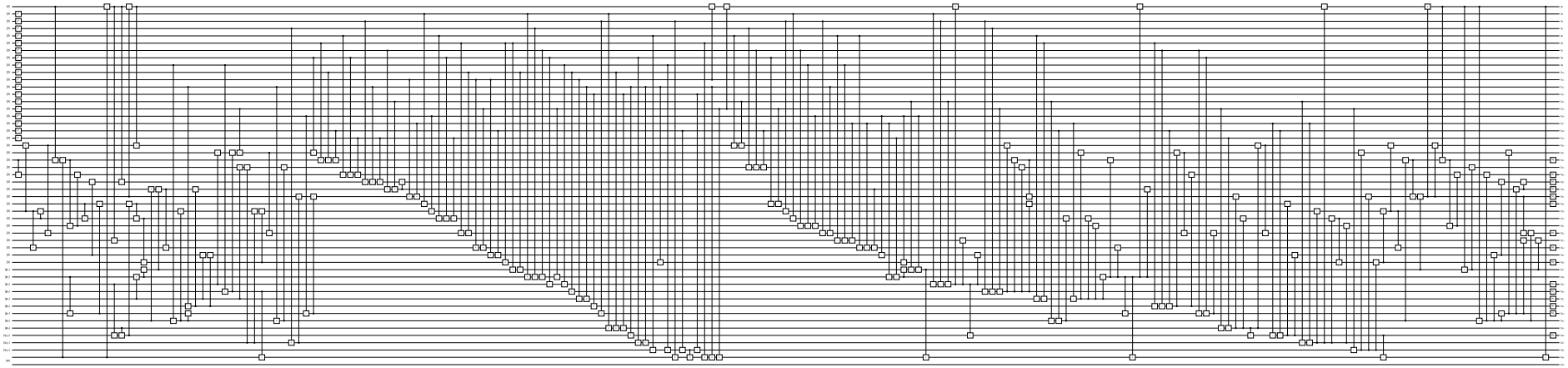


Figure 34: Encoder of the $[[49, 12; 1]]_3$ code of Example 9 obtained by the proposed encoding strategy followed by beam-search optimization: 226 ADD, 15 MUL, 18 DFT; two-qudit depth 45. A target labeled + (−) denotes ADD ($\mathrm{ADD}^\dagger$), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

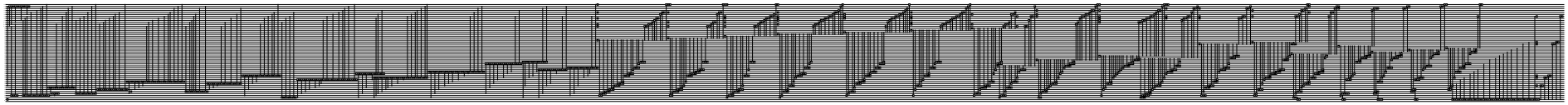


Figure 35: Encoder of the $[[49, 12; 1]]_3$ code of Example 9 obtained by the procedure of [NG]: 514 ADD, 531 MUL, 18 DFT, 1 SWAP; two-qudit depth 152. A target labeled + (−) denotes ADD ($\mathrm{ADD}^\dagger$), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

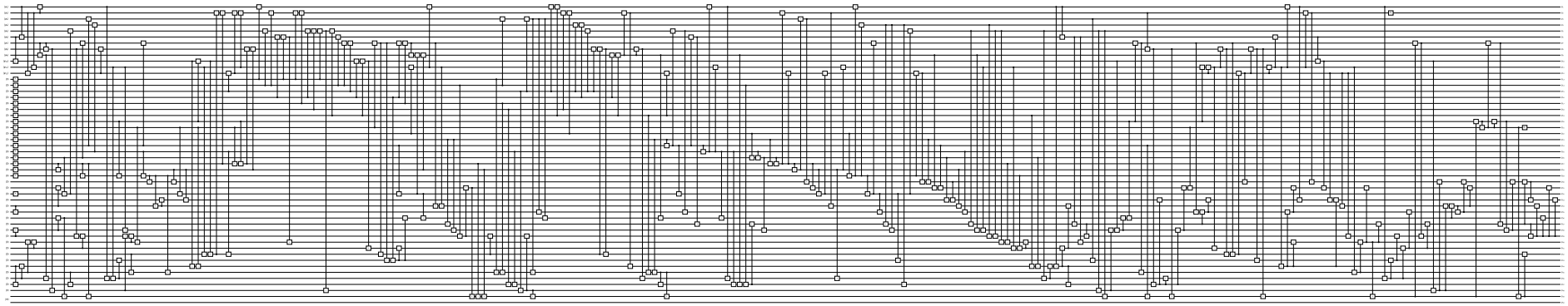


Figure 36: Encoder of the $[[49, 12; 1]]_3$ code of Example 9 obtained by the procedure of [NG] followed by beam-search optimization: 298 ADD, 2 MUL, 18 DFT; two-qudit depth 93. A target labeled $+$ ($-$) denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{3}} \sum_{x \in \mathbb{F}_3} |-x\rangle|x\rangle$; the receiver qudit q_{50} is not acted on.

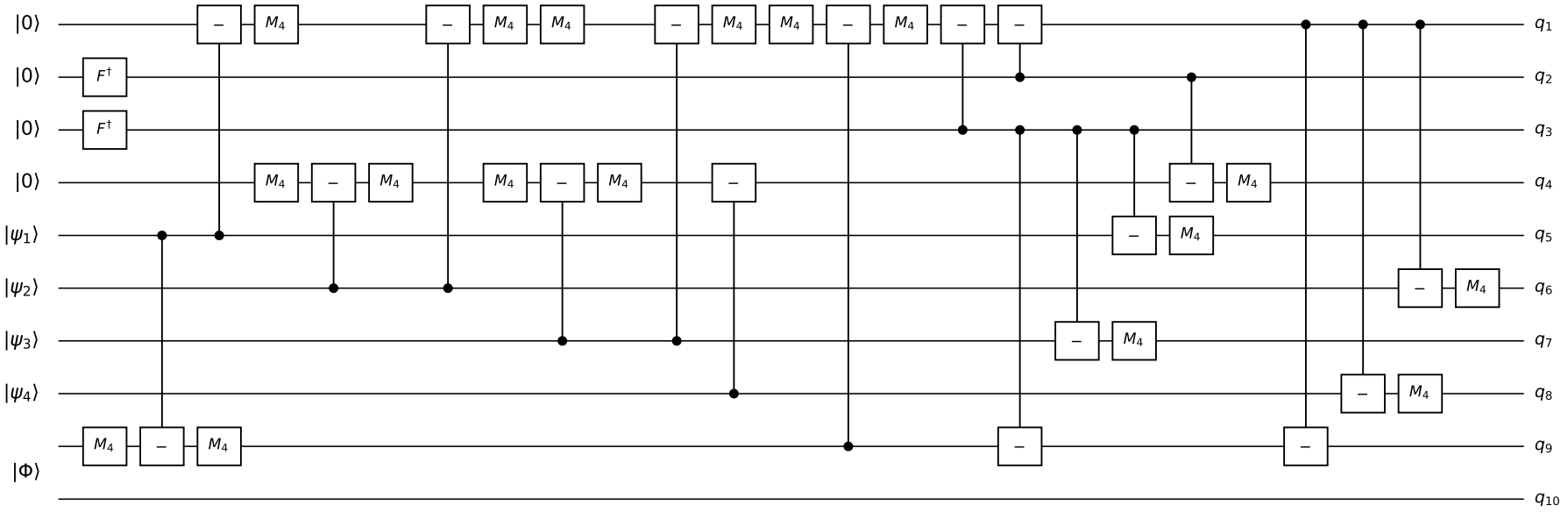


Figure 37: Encoder of the $[[9, 4; 1]]_5$ code of Example 10 obtained by the proposed encoding strategy: 17 ADD, 17 MUL, 2 DFT; two-qudit depth 10. A target labeled + (−) denotes ADD ($\text{ADD}^\dagger$), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{5}} \sum_{x \in \mathbb{F}_5} |-x\rangle|x\rangle$; the receiver qudit q_{10} is not acted on.

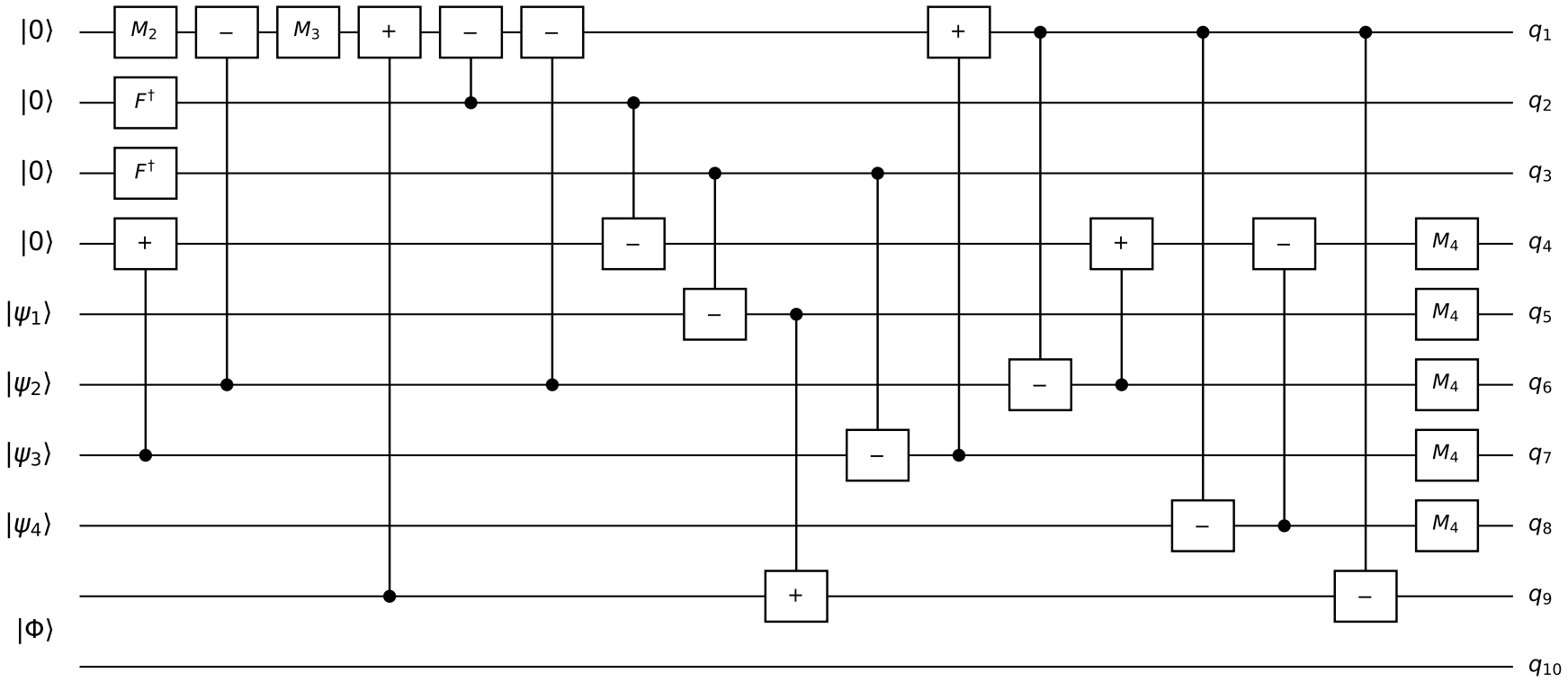


Figure 38: Encoder of the $[[9, 4; 1]]_5$ code of Example 10 obtained by the proposed encoding strategy followed by beam-search optimization: 15 ADD, 7 MUL, 2 DFT; two-qudit depth 8. A target labeled $+$ ($-$) denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{5}} \sum_{x \in \mathbb{F}_5} |-x\rangle|x\rangle$; the receiver qudit q_{10} is not acted on.

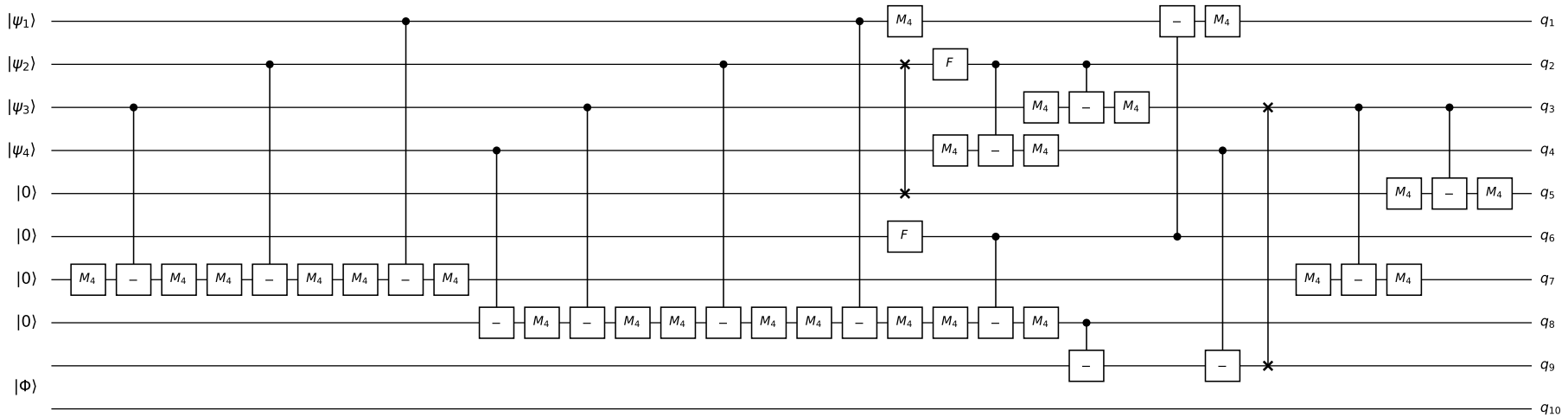


Figure 39: Encoder of the $[[9,4;1]]_5$ code of Example 10 obtained by the procedure of [NG]: 15 ADD, 24 MUL, 2 DFT, 2 SWAP; two-qudit depth 10. A target labeled $+$ $(-)$ denotes ADD $(\mathrm{ADD}^\dagger)$, $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{5}} \sum_{x \in \mathbb{F}_5} |-x\rangle|x\rangle$; the receiver qudit q_{10} is not acted on.

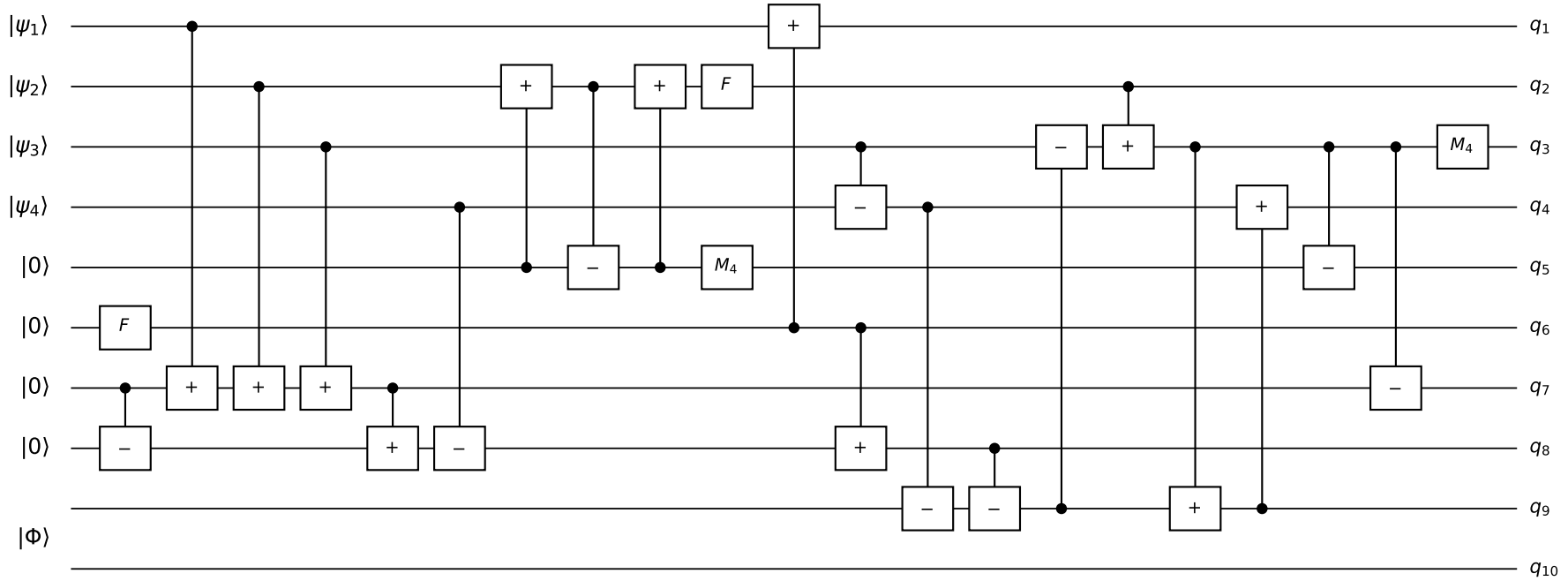


Figure 40: Encoder of the $[[9, 4; 1]]_5$ code of Example 10 obtained by the procedure of [NG] followed by beam-search optimization: 20 ADD, 2 MUL, 2 DFT; two-qudit depth 14. A target labeled + (−) denotes ADD (ADD†), $M_\gamma : |a\rangle \mapsto |\gamma a\rangle$, $F^\dagger$ is the inverse DFT, and $|\Phi\rangle = \frac{1}{\sqrt{5}} \sum_{x \in \mathbb{F}_5} |-x\rangle|x\rangle$; the receiver qudit q_{10} is not acted on.